

Method for Removing Background in LCC Videos

1. Introduction

These instructions outline the method for training and deploying a custom background removal model. It explains handpicking video frames as training data, augmenting the dataset using CloudSegNet, training a segmentation model with U-2-Net, and converting the trained model to ONNX format for usage with the Rembg library.

2. Dataset Preparation

Video can be converted into frames using the createFrames.py script. Input and output paths must be manually updated in the script. Picking the frames for training can be random or an iterative increase like one every 100 frames.

2A. Ground Truths

After choosing the frames to use, their ground truth counterparts must be manually created.

Ground Truths are a grayscale 1-layer image that displays to the computer what must be kept rather than left out.



Frame 0 of Run #1805 and Ground truth

To create the ground truths, softwares such as GIMP can be used. The process is simple and follows importing the frame then creating a transparent layer above the frame. Then the paint tool and fill tool can outline the cloud in white and background in black. Ensure the image color mode is set to greyscale and export as .png with the frame name.

Generally, 20-30 frames works well for most videos, but for better results more frames should be manually made. See results for references for comparable results.

3. Data Augmentation using CloudSegNet

Clone the CloudSegNet repo on github. To use the `create_aug_day.py` script in CloudSegNet, create a Conda environment using the provided `env.yaml` file. The github has additional information of the CloudSegNet program

Set the `NO_OF_IMAGES` parameter in `create_aug_day.py` to match the number of original frames. Input images should follow the naming format:

- Images: 0001.jpg, 0002.jpg, ..., 00NN.jpg

- Ground truth masks: 0001_GT.png, 0002_GT.png, ..., 00NN_GT.png

Running the script generates five augmented variants per image, increasing the dataset size by a factor of five.

4. Model Training using U-2-Net

Clone the U-2-Net repo on github. U-2-Net requires a different set of libraries and can be found in the requirements.txt

Prepare the directory structure in the U-2-Net root as follows:

```
train_data/
  DUTS/
    DUTS_TR/
      DUTS_TR/
        im_aug/
        gt_aug/
```

Copy the augmented images into im_aug and the corresponding masks into gt_aug. Update the save_frq variable in u2net_train.py to adjust model checkpoint frequency (e.g., 300 for quick testing). Train the model using the standard training script. To quickly test saved models, use u2net_test.py after placing test images in test_data/test_images. If the results seem washed out and nothing removed, check the dataset for miscorrelated images. If the dataset is correlated, then restarting the training script often generates good results without changing the dataset.

5. Model Conversion to ONNX

Once training completes, navigate to `saved_models/u2net/` and rename the desired `.pth` model file to `u2net.pth`. Run `export.py` to convert this model to ONNX format. Ensure the required packages listed in `export.py` are installed.

6. Deployment with Rembg

6.1 Command-Line Interface

To apply the trained model using the Rembg CLI:

```
rembg i -m u2net_custom -x '{"model_path": "~/.u2net/u2net.onnx"}' path/to/input.png  
path/to/output.png
```

Running the frames through the model multiple times can help refine segmentation output.

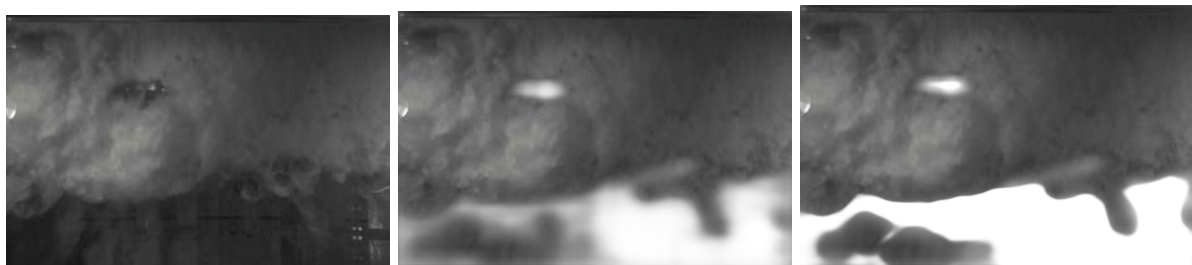
6.2 Python

Use `removeBackground.py` as an example to remove backgrounds of a folder of frames. Use `createVideo.py` to recompile processed frames into a video. Running the frames through the model multiple times can help refine segmentation output.

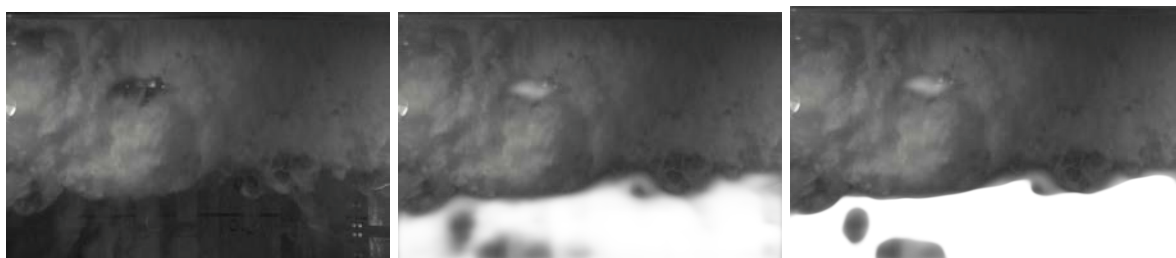
7. Results



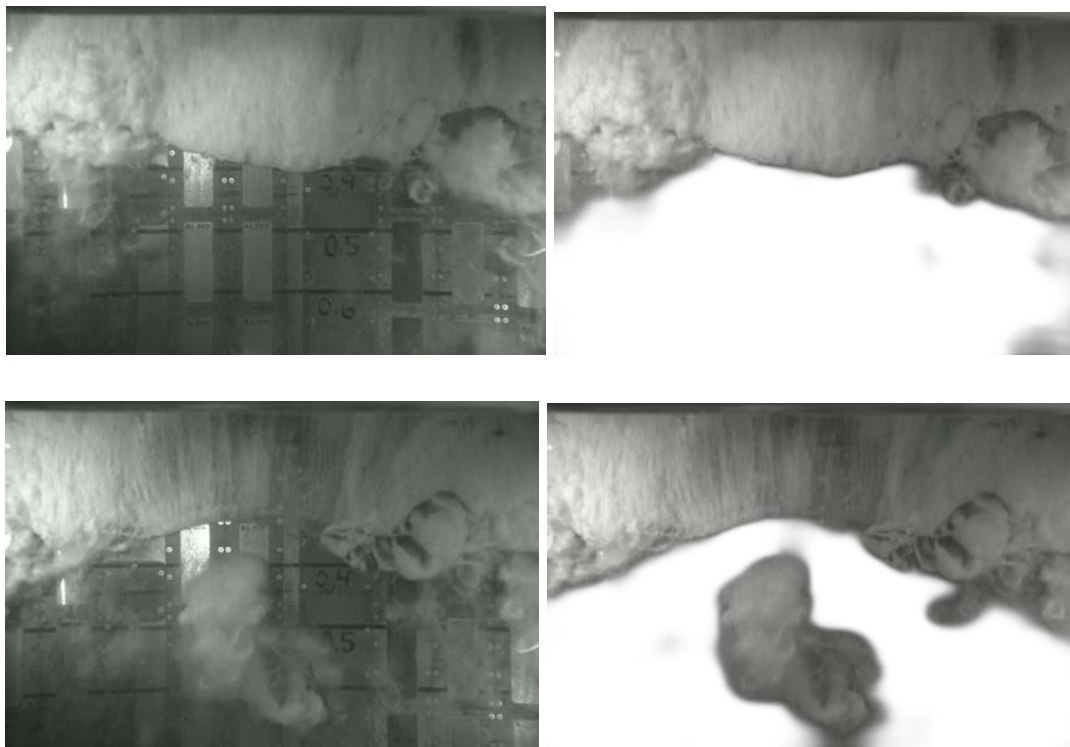
Run #1805 frame 3 - Given 50 frames of Run #1805 for 600 iterations



*Run #1451 frames 235 and 1160 - Given 31 frames of Run #1451 for 150 iterations with one
restart and cleaned with clean.onnx*

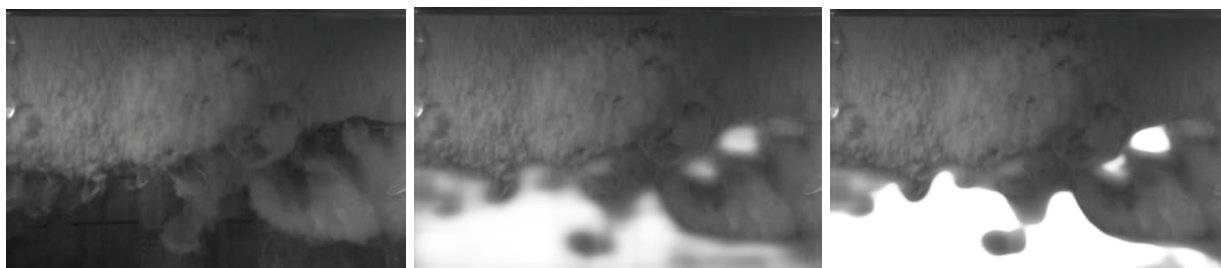


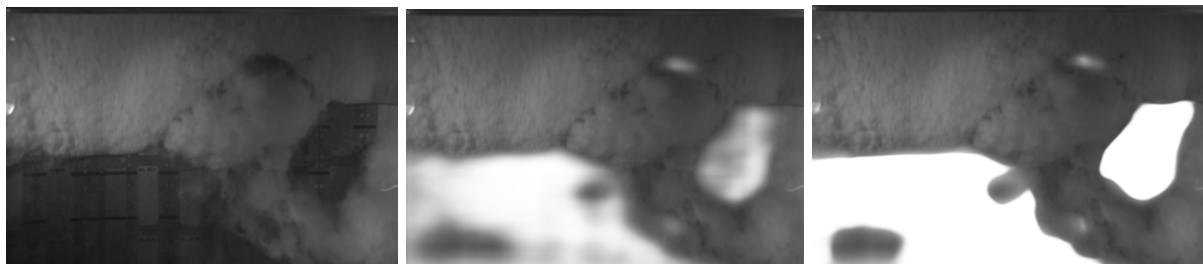
Run #1451 frame 1160 - Given 50 frames of Run #1451 for 150 iterations with one restart and cleaned with clean.onnx



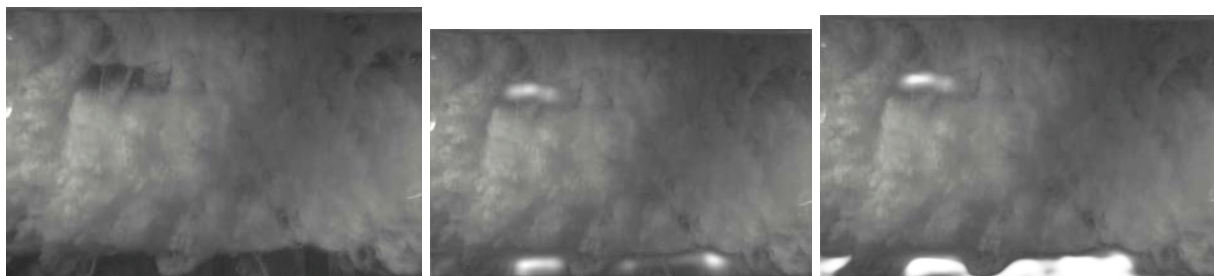
HighShedding Frames 2859 & 2930 - Given 30 frames 450 iterations Batch Size 15

For videos with similar chord lengths and light, models can be reused.





Run #1452 frame 733 and frame 963 - Used the saved model 30 frames of #1451



Run #1454 frame 1707 - Used the saved Model 50 frames of #1451 cleaned with clean.onnx

References

- Dev, S., Nautiyal, A., Lee, Y. H., & Winkler, S. (2019). CloudSegNet: A deep network for nychthemeron cloud image segmentation. *IEEE Geoscience and Remote Sensing Letters*, 16(12), 1814-1818
- Gatis, D. (2025). rembg (Version 2.0.66) [Computer software].
<https://github.com/danielgatis/rembg>
- Qin, X., Zhang, Z., Huang, C., Dehghan, M., Zaiane, O. R., & Jagersand, M. (2020). U2-Net: Going deeper with nested U-structure for salient object detection. *Pattern recognition*, 106, 107404.